Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/10/2025
PI Name:	Robinson, Lucy Ph.D.		
Project Title:	Investigating Lunar Stress and Parkinson's Disease Using an Alpha Synuclein Yeast Model		
Division Name:	Space Biology		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	71130	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-LEIA E.10. Lunar Explorer Instrument for Space Biology Applications
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 07/30/2024 per NSSC information (Ed., 1/4/24).		
Key Personnel Changes/Previous PI:	NOTE: Per the PI, collaborators "Runhua Shi, Ph.D. (Biostatistician; not involved in third year or NCE period) Jeffrey Chancellor, Ph.D. (LSU-Baton Rouge, not involved in third year or NCE period)", so were removed from the list of Co-Investigators for this reporting period (Ed., 2/22/25). NOTE: Per NASA Ames Research Center, the former Principal Investigator (Lynn Harrison, Ph.D., Louisiana State University) has moved to a new position and will no longer serve as PI on this project. A former Co-Investigator on the project will serve as the new PI. The new PI is Lucy Robinson, Ph.D. of Louisiana State University Health Sciences Center at Shreveport. (Ed., 3/7/23)		
COI Name (Institution):	Tatchell, Kelly Ph.D. (Louisiana State University System)		
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Task Description:	Astronauts will be exposed to partial gravity and deep space high energy radiation when living on the surface of the moon. Ionizing radiation increases oxidative stress in cells, and persistent oxidative stress in neurons from repeated exposure to low dose high energy radiation could increase the risk of developing neurodegenerative disease. Experiments are therefore needed to determine whether lunar living conditions increase toxic forms of proteins known to be involved in neurodegeneration. Alpha-synuclein (SYN) and mutations of this protein are implicated in Parkinson's disease. One such familial mutation is A53T-SYN, which increases aggregate formation and toxicity of SYN. Toxic forms of SYN can also form in cells under oxidative stress. The objective of this proposal is to determine whether lunar radiation or the stress of exposure to one sixth the Earth's gravity increases the toxicity of SYN. Two aims are proposed: aim 1 is to test the experiment in the Biosensor hardware to prepare for a possible lunar lander mission in the future and aim 2 is to use ground-based studies to determine whether SYN or A53T-SYN toxicity is increased by simulated lunar radiation or simulated lunar gravity. The Biosensor hardware has been tested for toxicity studies using yeast and the expression of SYN in Saccharomyces cerevisiae is an established model of Parkinson's disease. Low SYN levels in yeast are not toxic, but high levels induce apoptosis and Reactive Oxidative Species (ROS) accumulation. We will generate wild-type and RAD51 deleted Saccharomyces cerevisiae strains that can be induced to express SYN or A53T-SYN at low levels by red light. RAD51 deleted yeast are more radiosensitive than wild-type yeast and will increase the sensitivity of detection of SYN toxicity generated by radiation/oxidative stress exposure. Genes known to reduce SYN toxicity in yeast include the gene encoding manganese superoxide dismutase (SOD2), an enzyme that detoxifies superoxide radicals in mitochondria. We will therefore also develop strains that overexpress SOD2 in combination with SYN to determine whether SYN toxicity can be rescued by quenching superoxide radicals. The Biosensor hardware can determine toxicity by the Alamar Blue viability assay, and growth by culture turbidity. We will use these assays in aims 1 and 2 to determine SYN toxicity as well as colony-forming ability in the ground-based studies in aim 2. In aim 2, simulated lunar gravity will be achieved using the random positioning machine (RPM 2.0) at Kennedy Space Center and simulated lunar radiation exposure will be performed at Brookhaven. Dr. Chancellor, a team member, has developed techniques to recreate the polyenergetic radiation spectrum for ground-based studies of galactic cosmic rays, and will use measurements scheduled to occur in early 2022 on the lunar surface to guide exposure of the cultures at Brookhaven in aim 2. From the work in this proposal, we will be able to study SYN toxicity. We hypothesize that expressing SYN or A53T-SYN in yeast will increase toxicity to simulated lunar radiation or simulated lunar gravity, and toxicity will be greater in yeast expressing A53T-SYN. Toxicity will be decreased by overexpressing SOD2, an enzyme known to protect against SYN toxicity. Understanding how the stresses of living on the moon can enhance SYN toxicity is significant to humans thriving in deep space as extended space missions on the moon could enhance neurodegeneration in astronauts. Knowing the risks will allow NASA to establish countermeasures to protect astronauts on future deep space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This will be the first time that a yeast model of Parkinson's disease will be used to study the effect of lunar gravity and the lunar radiation environment on aggregation of a protein implicated in Parkinson's disease. This could have implications for risk assessment of neurodegeneration for astronauts living on the Moon. These experiments will provide the ground-based studies needed to justify performing experiments on the Moon. They will open up this field of research to other space biologists so the risk of neurodegeneration can be assessed in animals on the Moon. Without knowing how lunar living stress conditions alter brain physiology, it is not possible to design necessary countermeasures to keep our astronauts safe. Understanding the effects of alpha synuclein on irradiated cells will also have implications for radiotherapy patients that receive radiation to the brain. Experiments include initial tests with gamma radiation.
Task Progress:	Objective: The objective of this work was to prepare an experiment to determine whether stress from living on the lunar surface enhances the toxicity of the alpha-synuclein (SYN) protein, alteration of which is associated with neurodegeneration and Parkinson's disease, and if increased production of the anti-oxidant protein SOD2 could decrease such toxicity. Background: During spaceflight, astronauts are exposed to microgravity and radiation; two major stressors that increase oxidative stress. Living on the lunar surface exposes astronauts to 1/6th Earth's gravity and deep space high energy radiation, and little is known about the effects of these stress conditions on human cells. Since stress conditions of partial gravity and deep space radiation cannot be mimicked on the International Space Station (ISS), it is necessary to design and perform experiments on the lunar surface that are informative about potential human health risks. Ionizing radiation increases reactive oxygen species (ROS) in cells, and oxidative stress and DNA damage are implicated in neurodegenerative disease. An example is Parkinson's disease, where oxidative stress is a key player in the disease and can result in formation of aggregates of the SYN protein that accumulate and further exacerbate oxidative stress, leading to death of dopaminergic neurons. Astronauts living on the Moon could be at risk for developing neurodegenerative disease later in life due to a persistent oxidative stress in neurons from repeated exposure to low dose high linear energy transfer (LET) radiation and partial gravity. The expression of the SYN protein in the budding yeast Saccharomyces cerevisiae is an established model of Parkinson's disease. Low SYN protein levels in yeast are not toxic, but high levels induce ROS accumulation and eventually cell death. Loss of manganese superoxide dismutase (SOD2), an enzyme that detoxifies superoxide radicals in mitochondria, enhances the toxicity of high levels of SYN protein. Hypotheses: Expressing SYN or A53T-SYN (a familial mutation found in Parkinson's patients) protein in yeast will increase toxicity of the simulated lunar radiation environment or lunar gravity, and cell death will be greater in yeast expressing A53T-SYN. Toxicity will be decreased by overexpressing SOD2. AIM 1: Test SYN and A53T-SYN toxicity in combination with SOD2. AIM 2: Test toxicity of simulated deep space radiation and simulated lunar gravity on yeast expressing SYN or A53T-SYN protein in combination with SOD2 overexpression. Overall Progress Although the reagents and strains we planned to use in the two aims were eventually generated, neither Aim was completed due to technical challenges and unforeseen circumstances. Most progress was made for Aim 1. We

constructed all yeast strains needed for our experiments in a genetic background that can tolerate stasis in drying conditions (desiccation) that is necessary for testing in simulated space conditions and constructed plasmids that allow inducible expression of the SYN protein in our strains, which comprised a major part of Aim 1. Preliminary data generated with these strains were consistent with the hypothesis underlying Aim 2, but unexpected circumstances prevented the thorough analysis we proposed and intended.

Originally, we proposed to use a system to induce yeast to express the SYN protein upon exposure to red light (wavelength 600-660). This would have been an excellent system for use in the proposed experiments to simulate lunar gravity and deep space irradiation, as it would not require any change to growth conditions except addition of the red light. We engineered the expression system and yeast strains to produce the required components. However, our plans to use the red light induction system had to be changed after we observed ill effects of the expression system alone on cell growth and cell division in dark conditions, and these were exacerbated by treatment with red light. Therefore, we abandoned the use of this system for inducible expression of the SYN protein and turned to galactose-inducible (GAL) expression, a reliable system for inducible expression in yeast. We constructed a low copy plasmid allowing galactose-inducible expression of SYN protein in any of our engineered yeast strains and confirmed using an anti-SYN antibody that the protein is not detectable in the absence of galactose but is abundant within three hours of galactose addition to culture media, but were unable to complete the analysis of all yeast strains we generated.

Our initial experiment to test the effect of increasing anti-oxidant protein production on radiation survival of yeast strains provided results that support our hypothesis that SOD2 overexpresssion can protect cells from loss of viability due to gamma irradiation. We intended and prepared to carry out similar experiments with our engineered strains with and without expression of SYN, but unforeseen circumstances prevented completion of these experiments.

Bibliography Type: Description: (Last Updated:)